

وَمَا أُوتِيتُمْ مِنَ الْعِلْمِ إِلَّا قَلِيلًا



The Simplest Filter Ever

“Simplicity Always Wins”

Outline

- ❑ Introduction & Overview
- ❑ Workflow & Design Procedure
 - Filter Response & Filter Topology Selection
 - Behavioral Model
 - Design methodology
 - Core Amplifier
 - Output Stage
 - CMFB Network
- ❑ Automation: Obstacles & Workarounds
- ❑ Calibration Via VerilogA: Challenges
- ❑ Summary of Results
- ❑ Conclusions

Introduction & Overview

Introduction & Overview

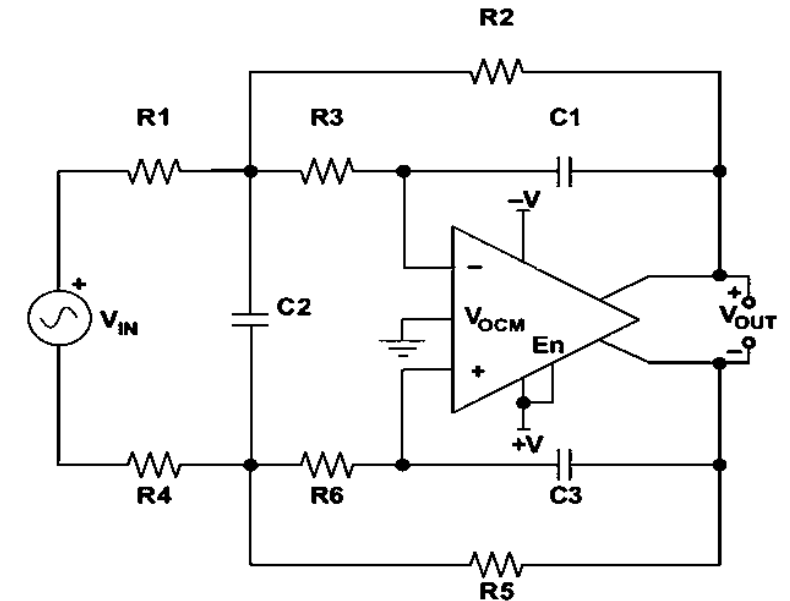
- ❑ We were assigned to design a fully differential RC active filter with a certain set of constraints and specifications.
- ❑ The goal was set to solve the problem using the simplest approach.
- ❑ This is evident in a multitude of aspects of our design.
 - Specs satisfaction via one – OpAmp filter topology.
 - The output stage.
 - The two – transistor CMFB network.

Workflow & Design Procedure

Workflow & Design Procedure

Filter Response & Filter Topology Selection

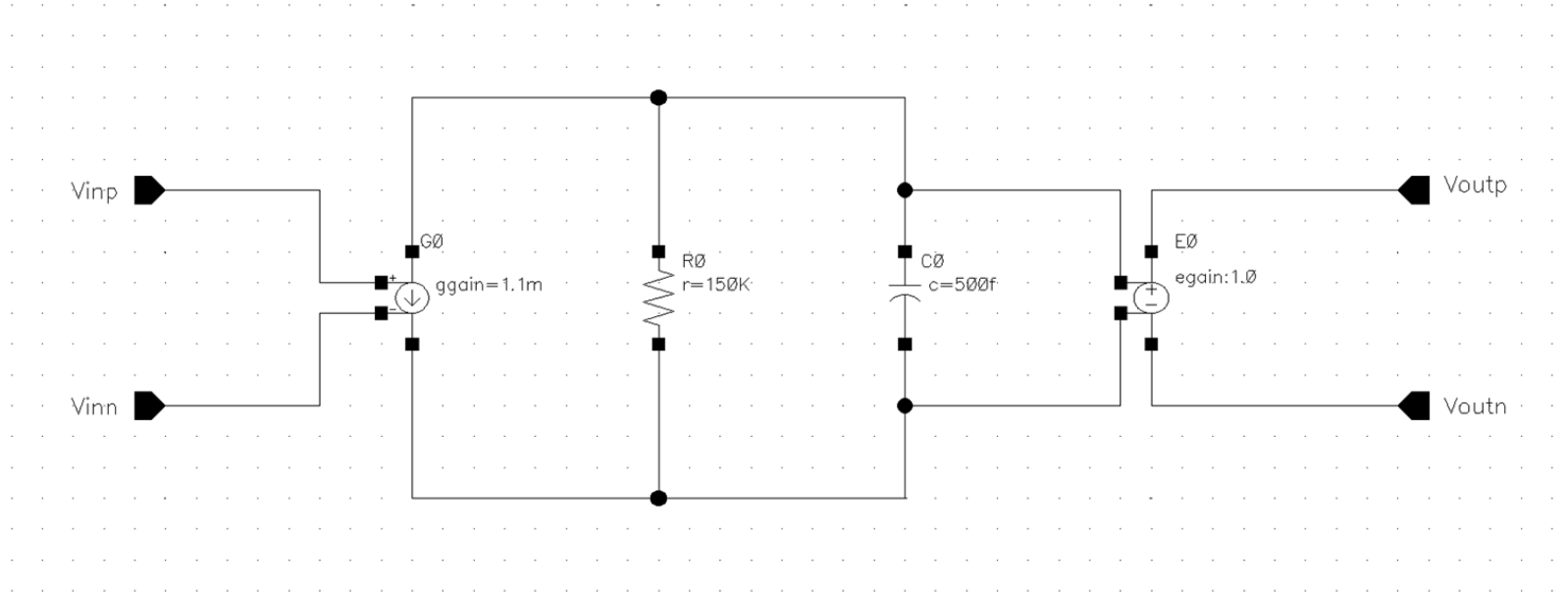
- ❑ First, Filter theory literature was reviewed.
 - Bessel response, for example, was considered, but its stopband CCs were not enough.
- ❑ The final decision has been made to use a “heterogeneous” response, which gives a linear phase response with 0.5° error, satisfying both group delay variation and stopband attenuation requirements.
- ❑ As simplicity was the dominant mindset while working, The goal was to choose a realization having the smallest possible number of Op Amps.
- ❑ Multiple feedback topology was the magic stick.



Workflow & Design Procedure

Behavioral Model

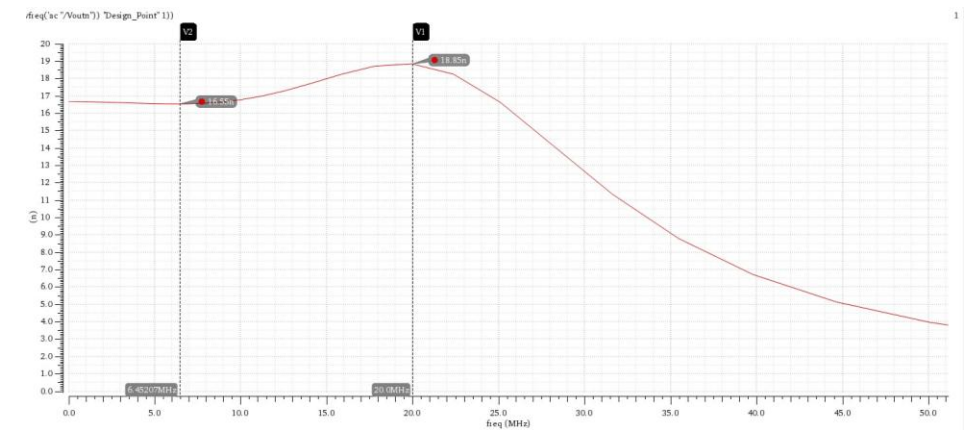
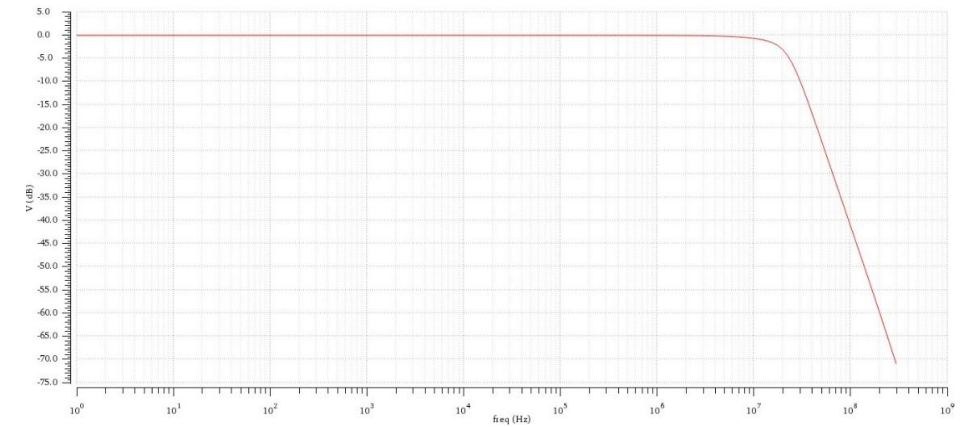
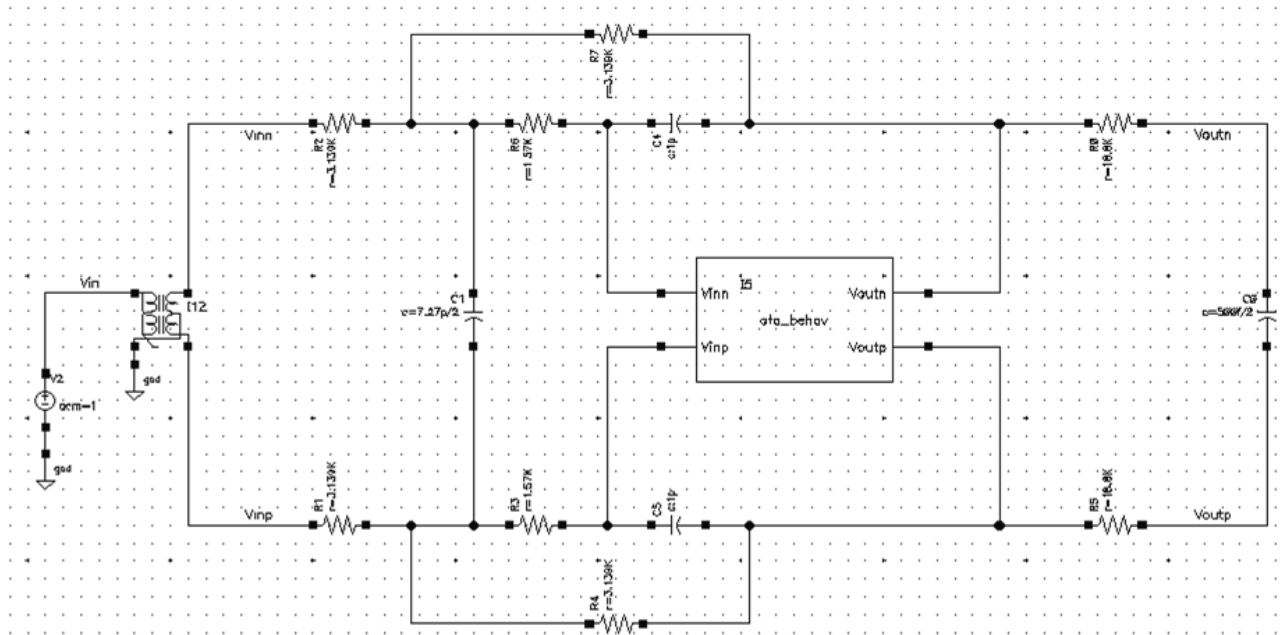
- ❑ First, a high-level-of-abstraction behavioral model was constructed to help estimating design parameters.
 - Testing chosen filter response & topology.
 - Active stage input pair transconductance, power consumption, noise budget, etc...



Workflow & Design Procedure

Behavioral Model (CONT'D)

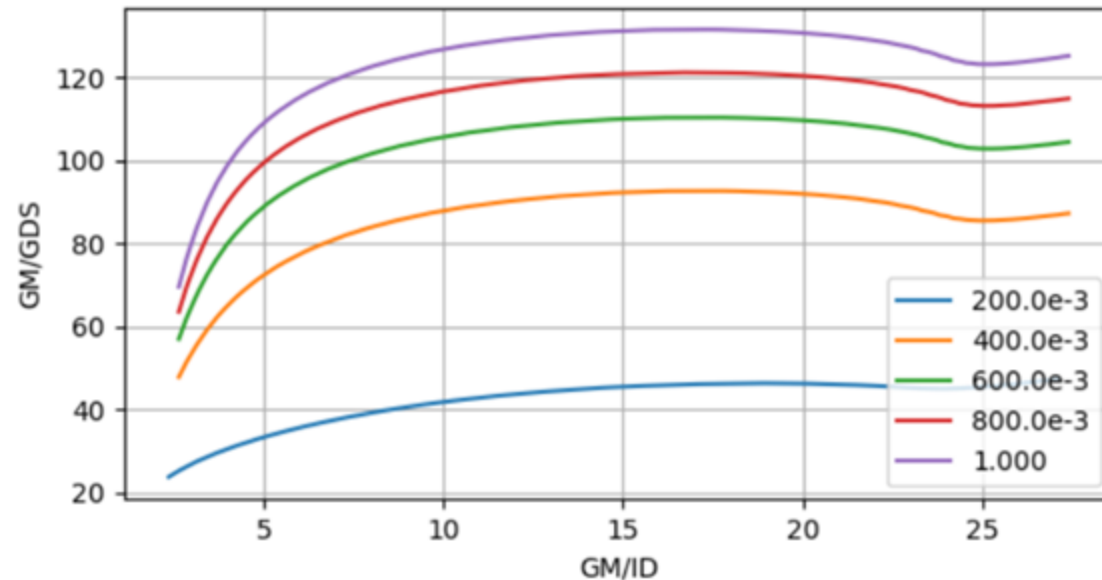
- ❑ Simulations were run and initial numbers were extracted...



Workflow & Design Procedure

Design methodology

- ❑ gm/ID design methodology was the main procedure followed.
 - gm/ID allows taking informed decisions based on precomputed design charts.
 - For example, device length is precisely calculated, not roughly estimated.
 - Whenever specs are not met via gm/ID, fine tuning comes to the surface.



Workflow & Design Procedure

❑ Tow-Thomas (TT) Biquad:

To achieve max peaking of 1 dB with the required bandwidth (20 MHz) and stop band attenuation.

$$GBW_{OP-AMP} \geq 336 \text{ MHz} \quad (\text{Analytically})$$

Pros:

- Gain, quality factor (Q) and cutoff frequency can be adjusted independently.
- PVT variations are minimized.

Cons:

- $GBW_{OP-AMP} \geq 1.2 \text{ GHz}$ to maintain the required group delay variation.
- Power Consumption, and Area.

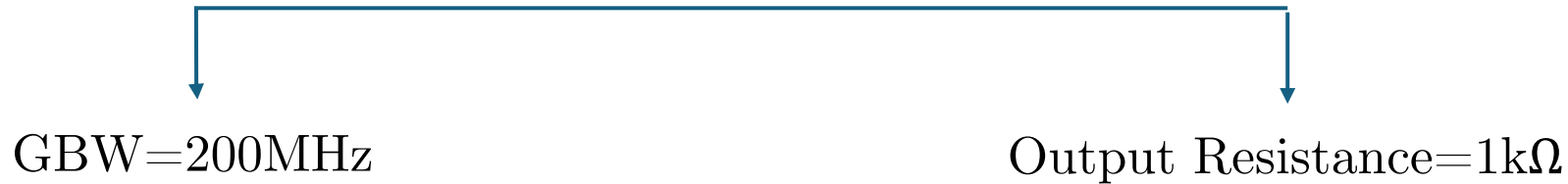
❑ Sallen-key:

Cons:

- The filter is **not easily to be tuned** since the interaction of component values on F_0 and Q.
- **Inductive response:** When the amplifier's output impedance is greater than (R1), the feedback looks **inductive** and the response increases at a rate of 20 dB/decade.

Workflow & Design Procedure

Behavioral Model Insights (running sweeps)



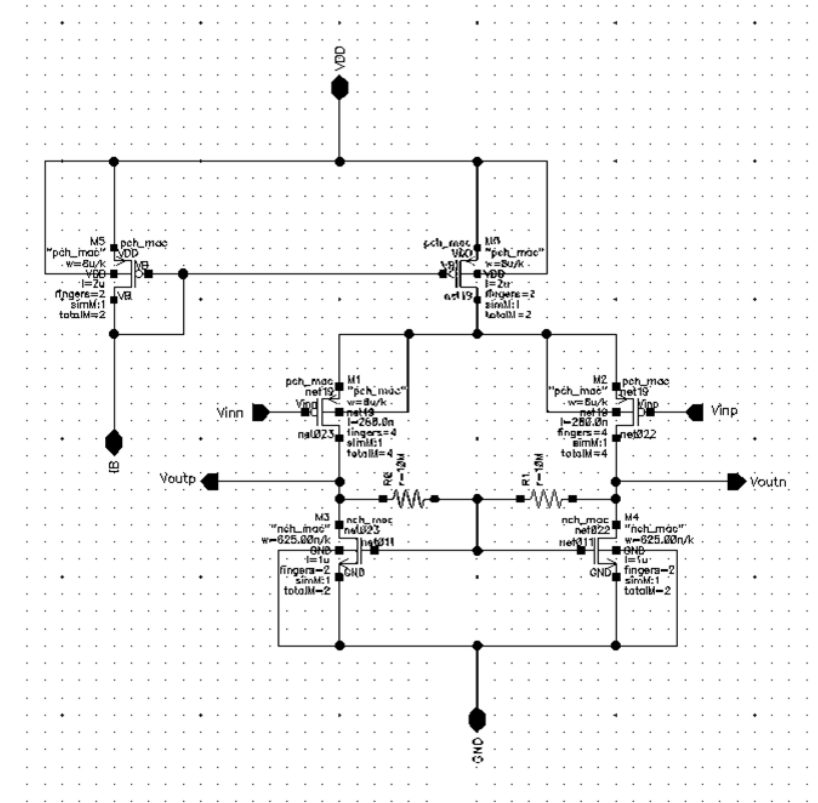
Parameter	Result
BW	20.01 MHz (required 20 MHz)
Out-of-band attenuation (@ 200MHz)	-59.55 dB (required -55 dB)
Max in-band group delay	2.18 nsec (required 3 nsec)

Workflow & Design Procedure

Core Amplifier (Fully differential 5T OTA)

- ❑ **PMOS Input Pair:** Lower flicker noise (Dominant in the band of operation).
- ❑ **Dummy CMFB network:** infinite resistances

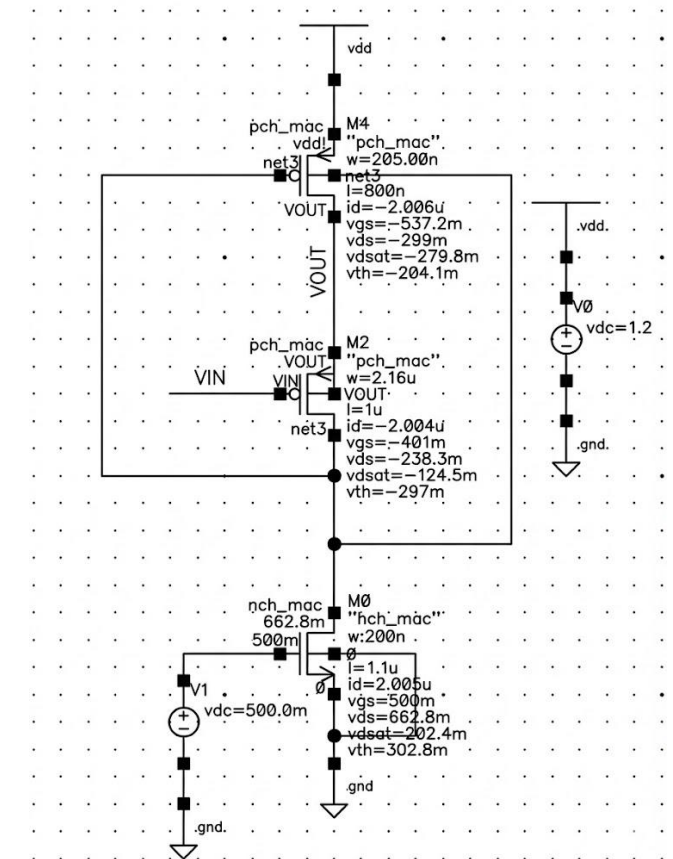
Spec	Result
Gain	29.27 dB (≈ 29.07)
GBW	2.432 GHz
Power Consumption	19 μ W
Output Impedance	Very Large (so, buffer is a must)
Input Range	300mV : 750mV (450mV)



Workflow & Design Procedure

Output Stage: Flipped Voltage Follower (FVF)

- ❑ Target: Output resistance $\leq 1k\Omega$
- ❑ FVF (Flipped Voltage Follower):
 - suffers a small Input-Output Swing.
 - For CMOS technologies with $V_T \sim 0.4V$
→ Typically, $V_{in(p-p)} < 300\text{ mV}^*$
 - A **200 mV** input swing was achieved successfully.



(*) Ramirez-Angulo, J. *et al.* (2013). The Flipped Voltage Follower: Theory and Applications. In: Fakhfakh, M., Tlelo-Cuautle, E., Castro-Lopez, R. (eds) Analog/RF and Mixed-Signal Circuit Systematic Design. Lecture Notes in Electrical Engineering, vol 233. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-36329-0_12

Workflow & Design Procedure

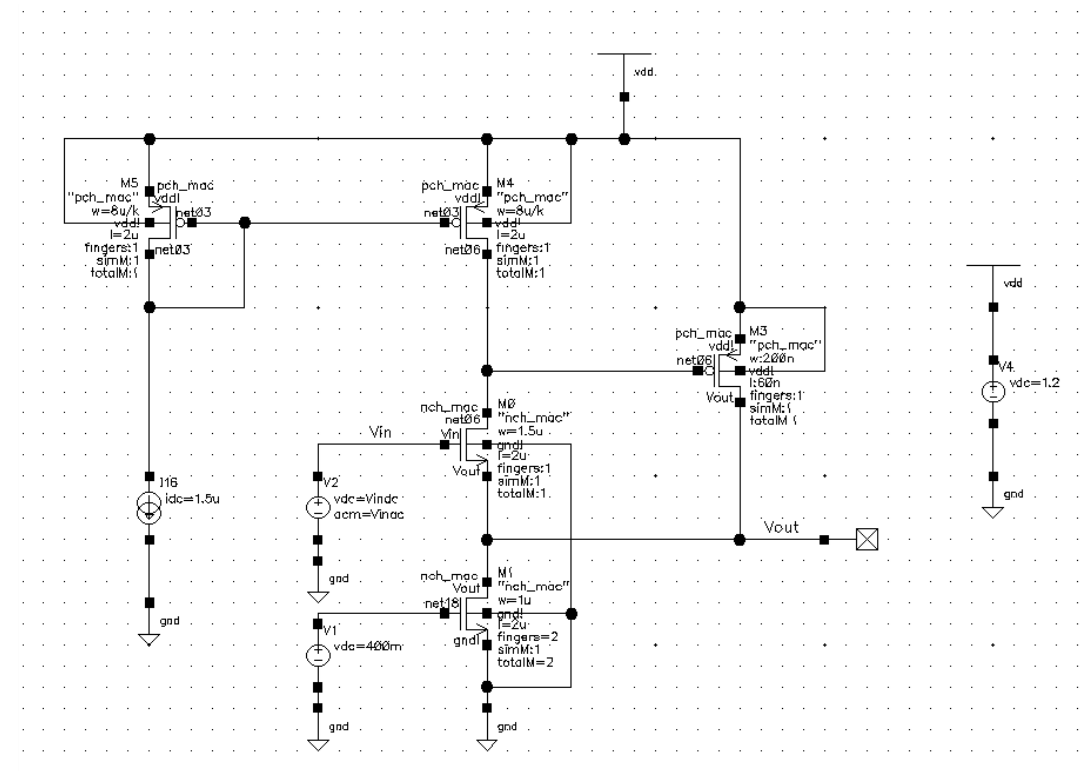
Output Stage: Super Source Follower (SSF)

❑ SSF (Super Source Follower):

- DC gain = -0.0834 dB
- Peaking of (2.755 dB) (New pole)
- DC Output resistance: $7.8 \text{ k}\Omega$
- Input Range: $(0.18:0.59) \text{ V}$ (410 mV_{pp})
- Power Consumption: $9.6 \mu\text{W}$

❑ Solution:

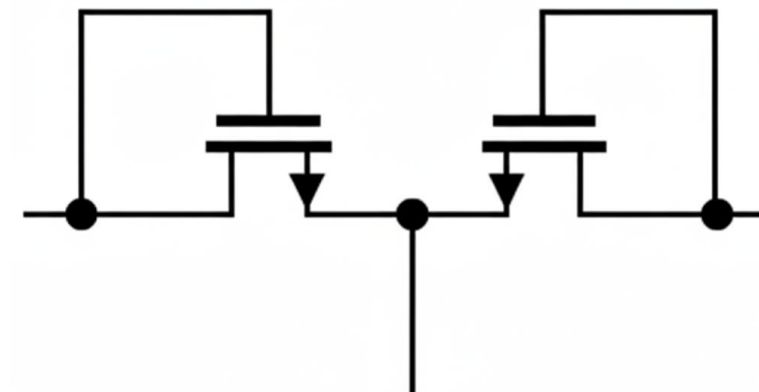
- The well-designed fully diff OTA in a unity negative feedback loop.



Workflow & Design Procedure

CMFB Network

- ❑ The simplest CMFB network comprises only sensing resistors + a short circuit.
 - Very large resistors are needed.
 - Huge area that isn't practical.
- ❑ A collection of transistors can be used to realize almost any RLC network.
 - A transistor can be diode – connected to realize a resistor.
 - Required large resistors are obtained.
 - But with practical area that can be easily laid out.



Automation: Obstacles & Workarounds

Automation: Obstacles & Workarounds

The Need for Automation in Analog IC Design

- ❑ Problem: Manual design is slow, and repetitive, especially with changing specifications or the technology used.
- ❑ Solution: Design automation frameworks like BAG, or simply using Python or MATLAB.

Automation: Obstacles & Workarounds

Why BAG?

- ❑ Parameterized Design: Creates Python-based generators, not just static designs.
- ❑ Technology Independence: A single generator can target multiple process technologies (e.g., 65nm, 40nm).
- ❑ Full Flow Integration: Manages schematic, layout, and verification in one environment.
- ❑ Reduces Time-to-Market: Drastically speeds up redesign cycles.

Automation: Obstacles & Workarounds

The Journey: A Dive into BAG3 Framework Setup

❑ Objective: To set up the BAG3 environment on Debian 8 VM.

❑ A Three-Phase Troubleshooting Journey:

➤ Phase 1: No Internet Connection

- Problem: The VM was completely offline. Basic commands like ping failed with Network is unreachable.
- Solution: Manually reconfigured the VM's network to get it online

➤ Phase 2: Fixing the Old System

- Problem: The OS was too old to download any software. Its software sources were dead and its security was outdated.
- Solution: Manually redirected the system to historical archives and bypassed all security errors to install basic tools like git and curl.

Automation: Obstacles & Workarounds

The Journey: A Dive into BAG3 Framework Setup (CONT'D)

➤ Phase 3: The Unfixable "GLIBC Wall"

- Problem: After finally installing the basic tools, the installation of Miniconda (a core BAG dependency) failed.
- Root Cause: A fundamental conflict between the old OS libraries (GLIBC 2.19) and the new tool requirements (GLIBC > 2.28).
- Conclusion: This is a fundamental incompatibility. The OS is simply too old for modern tools.

❑ We have tried to install BAG on CentOS but it didn't work too.

Automation: Obstacles & Workarounds

The Pivot: A Python-Based Generator

- ❑ Since the framework itself was inaccessible, we focused on the core principle: automating the design procedure.
 - Objective: Create a Python script that acts as a "design generator" for our 3rd-order Bessel filter.
 - Input: High-level specifications (e.g., $f_c = 40\text{MHz}$).
 - Process:
 1. Calculates passive component values (Resistors, Capacitors) based on filter theory.
 2. Determines required OTA performance.
 3. Sizes the transistors for the OTA.
 - Output: A complete list of component values and transistor dimensions (W/L).
- ❑ This approach mimics the procedural nature of a BAG generator.

Automation: Obstacles & Workarounds

Code Implementation: Methodology & Justification

❑ OTA Sizing Methodology

- The script uses the Square-Law model for initial transistor sizing.

❑ Justifications

- First-Order Approximation: The Square-Law model provides an excellent first-order estimate for transistor dimensions, ideal for an initial, automated design script. It's simple, fast, and captures the fundamental device physics.
- We started with a simplified model for initial sizing before moving to more complex models (like gm/ID) or simulation-based optimization in a real-world flow.
- Focus on Automation: The primary goal here is to demonstrate the automation flow itself, not to achieve perfect results at this stage. The model can be easily swapped for a gm/ID lookup table in a future iteration
- The chosen $V_{OV} = \text{almost } 0.2V$ ensures operation in strong inversion, where the Square-Law model provides a valid and robust first-order approximation for initial, automated sizing.

Automation: Obstacles & Workarounds

Modifications & Results

Parameter	Mod #1	Mod #2	Mod #3
Changed Specification	$f_c = 40 \text{ MHz}$	46 dBc out-of-band attenuation at 200 MHz	DR = 70 dB
Passive Components (2nd-Order MFB Section)	$R_1, R_2 = 1569.54 \text{ } \Omega$ $R_3 = 784.77 \text{ } \Omega$ $C_1 = 1 \text{ pF}$ $C_2 = 7.275 \text{ pF}$	$R_1, R_2 = 3139.09 \text{ } \Omega$ $R_3 = 1569.54 \text{ } \Omega$ $C_1 = 1 \text{ pF}$ $C_2 = 7.275 \text{ pF}$	$R_1, R_2 = 3139.09 \text{ } \Omega$ $R_3 = 1569.54 \text{ } \Omega$ $C_1 = 1 \text{ pF}$ $C_2 = 7.275 \text{ pF}$
Passive Components (1st-Order RC Section)	$R = 4818.79 \text{ } \Omega$ $C = 1 \text{ pF}$	$R = 9637.58 \text{ } \Omega$ $C = 1 \text{ pF}$	$R = 9637.58 \text{ } \Omega$ $C = 1 \text{ pF}$
Transistor Dimensions	Input pair: $I_d = 9.01 \text{ } \mu\text{A}$ $W = 0.90 \text{ } \mu\text{m}$ $L = 280 \text{ nm}$ Active load: $I_d = 9.01 \text{ } \mu\text{A}$ $W = 1.02 \text{ } \mu\text{m}$ $L = 1000 \text{ nm}$ Current mirror: $I_d = 18.01 \text{ } \mu\text{A}$ $W = 12.87 \text{ } \mu\text{m}$ $L = 2000 \text{ nm}$	Input pair: $I_d = 4.50 \text{ } \mu\text{A}$ $W = 0.45 \text{ } \mu\text{m}$ $L = 280 \text{ nm}$ Active load: $I_d = 4.50 \text{ } \mu\text{A}$ $W = 0.51 \text{ } \mu\text{m}$ $L = 1000 \text{ nm}$ Current mirror: $I_d = 9.01 \text{ } \mu\text{A}$ $W = 6.43 \text{ } \mu\text{m}$ $L = 2000 \text{ nm}$	Input pair: $I_d = 6.75 \text{ } \mu\text{A}$ $W = 0.68 \text{ } \mu\text{m}$ $L = 280 \text{ nm}$ Active load: $I_d = 6.75 \text{ } \mu\text{A}$ $W = 0.77 \text{ } \mu\text{m}$ $L = 1000 \text{ nm}$ Current mirror: $I_d = 13.51 \text{ } \mu\text{A}$ $W = 9.65 \text{ } \mu\text{m}$ $L = 2000 \text{ nm}$

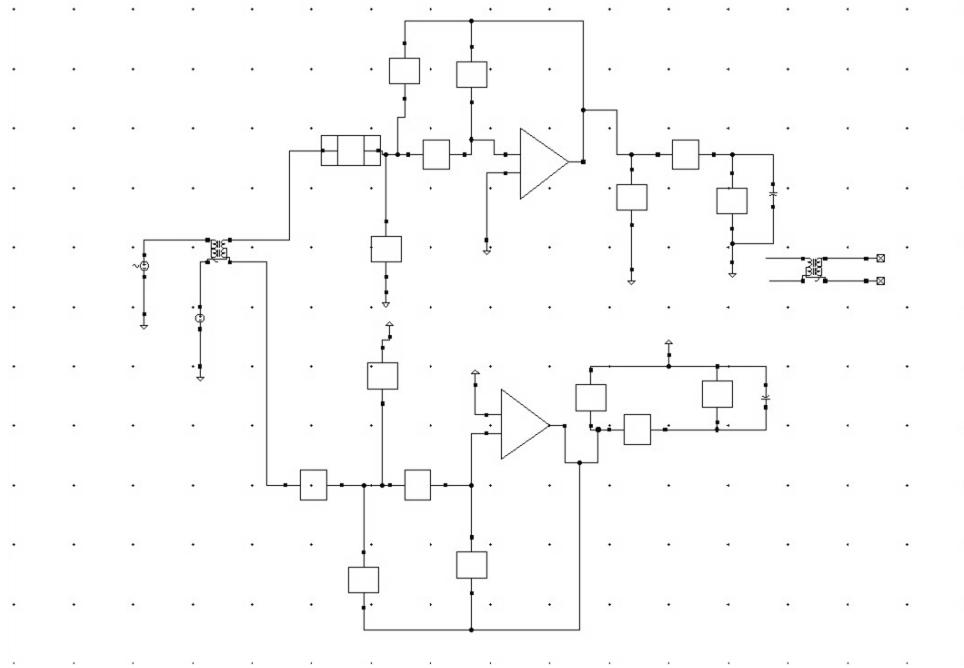
Calibration Via VerilogA: Challenges

Calibration Via VerilogA: Challenges

- ❑ PVT variations, if not considered properly, can deteriorate the circuit performance severely.
 - Calibration schemes are integral parts in IC technology.
- ❑ The problem statement required an emulation of a calibration scheme via a VerilogA code.
- ❑ An OpAmp, a resistance and a capacitance model was written.
 - Sub – blocks were connected to emulate the transistor – level circuit.
 - The performance were tested and compared to transistor – level performance.
 - Various mismatches were introduced to investigate their effect on the performance.
 - Unfortunately, time was not sufficient to complete this work.

Calibration Via VerilogA: Challenges (CONT'D)

❑ Following figures illustrates the performance variations Vs introduced mismatches.



Test	Output	Nominal	Spec	Weight	Pass/Fail
Si_project:verilog_test_becnch:1	/VOP				
Si_project:verilog_test_becnch:1	/VON				
Si_project:verilog_test_becnch:1	/VOUT				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOP"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOP") 3 "low")	20.01M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOP"))				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOUT"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOUT") 3 "lo...	20.01M			
Si_project:verilog_test_becnch:1	gainBwProd(VF("/VOUT"))	17.19M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOUT"))				

Test	Output	Nominal	Spec	Weight	Pass/Fail
Si_project:verilog_test_becnch:1	/VOP				
Si_project:verilog_test_becnch:1	/VON				
Si_project:verilog_test_becnch:1	/VOUT				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOP"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOP") 3 "low")	20.27M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOP"))				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOUT"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOUT") 3 "lo...	23.12M			
Si_project:verilog_test_becnch:1	gainBwProd(VF("/VOUT"))	15.91M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOUT"))				

Summary of Results

Summary of Results

□ Following is a summary of achieved specs under nominal operating conditions.

Spec	Required	Achieved
Gain & BW	0dB & 20MHz	0dB & 20MHz :)
Out-of-band attenuation @200MHz	55dBc	$\approx$ 60dB :)
Max in band group delay	3nsec	$\approx$ 2.2nsec :)
Dynamic range	50dB	49.755dB
THD	1%	3.9% :(
Power consumption	Minimum	85 μ W :)

Summary of Results (CONT'D)

□ Following is a summary of achieved specs across *supply voltage variation*.

Fractional Variation ($\Delta V_{DD} = \pm 10\%$)	
Spec	Fractional Variation
BW	2.8%
Out-of-Band Attenuation @200MHz	2360ppm
Max In Band Group Delay	0.15%
THD	1.33%
PM	2.87%
DR	$\approx 0\%$

Summary of Results (CONT'D)

□ Following is a summary of achieved specs across *temperature corners*.

Spec	-40°C	27°C	125°C
Out-of-Band Attenuation @200MHz	-70.8dB	-64.8dB	-59.5dB
Max In Band Group Delay	1.4nsec	1.7nsec	2.5nsec
THD %	3.707%	7.15%	32%
RMS noise of CL Op Amp	$172\mu V_{rms}$	$204\mu V_{rms}$	$265\mu V_{rms}$

Conclusions

- ❑ This work is a result of two months of hard working and collaboration between three senior ECE students and one ECE graduate (we're just four 😊)
- ❑ Very special thanks for our mentors, [Eng. Mohammad Magdy](#) and [Eng. Gehad Magdy](#) for their invaluable guidance and assistance throughout all the work phases.

Meet Our Team

- ❑ [Ahmed Raafat Ibrahim Zidan](#)
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Graduated at ECE department, Sohag University

Ready for Questions :)

Thanks ^^